REVIEW QUESTIONS

1.What is an IMU?
IMU stands for Inertial Measurement Unit. It is a motion sensing device that combines the data from a gyroscope and an accelerometer together. Two types of IMU based on DOFs.

2.What is yaw, and how does a 9-DoF IMU reduce yaw drift?
Yaw is rotation about the Z axis. It is basically a left and right movement.
6Dof IMU has an accelerometer and gyroscope only. Yaw is provided by the gyroscope in a 6DOF IMU. 9DOF IMU has an accelerometer, gyroscope and a magnetometer. It senses magnetic north. Yaw is provided by the magnetometer only in a 9 DOF IMU.

3.What do you mean by /etc/netplan?
This is the configuration directory used by Netplan on Ubuntu systems to store the YAML files containing networking configuration information such as IP addresses, DNS, and routing.

Explain what /etc/netplan does in a Linux distro
It contains YAML files that contain instructions about how the interfaces should be set up on the network. Once the command netplan apply is executed, Netplan will read the files and create configuration files for the backend that is being used either systemd-networkd (used in server systems) or NetworkManager (used in desktop systems).

Is a cloud service a virtual machine? Are all cloud services VMs?
The latter represents just a particular type of the former.

First of all, a virtual machine (VM) is a computer hardware emulator implemented via software. It possesses its own operating system installed as a guest, its own processor and memory, storage and networking and operates like an ordinary standalone computer from the viewpoint of the user. For example, it can be AWS EC2, Azure Virtual Machines or Google Compute Engine.

On the contrary, a cloud service is a more general notion covering different types of computer resources delivered over the internet from the cloud provider. Therefore, cloud services include VMs but also many other services which do not have anything to do with managing virtual machines.

One of the crucial differences between these two concepts lies in the level of abstraction a user controls and is responsible for the most issues in case of VM while the level of abstraction is extremely high in case of, for instance, SaaS or serverless. That is why, one may conclude that every VM is a cloud service but not vice versa.

- Advantages of Docker
  Resource-efficient: containers reuse the OS kernel from the host operating system rather than emulate a complete operating system, thus consuming significantly fewer resources compared to VMs.

- Consistency: the package with the application and all its dependencies in the container guarantees consistent behavior across development, testing, and production environments, thus preventing issues such as "works on my machine."
- Portability: a Docker image will run consistently on any host with Docker, independent of the underlying operating system.

- Isolation: each container runs separately, hence avoiding any dependency conflicts between applications or services running in the same environment.

- Quick start: containers usually start within seconds, while VMs require several minutes to boot.
- Efficient resource consumption: many containers can run on one host because there is no need for a guest OS on each of them unlike in the case of VMs.

- Scalability: containers start and stop quickly which together with orchestration makes scaling up/down of applications possible.

- Version control of environments: the Dockerfiles and Docker images themselves can be version-controlled allowing for rollback in case of issues.